

Software engineer specializing in distributed runtimes, AI infrastructure, programming languages, and developer tooling.

Skills

Languages: Rust, Python, TypeScript, JavaScript, C++, OCaml, Scala.

Backend & Systems: Compilers, type systems, runtimes, distributed systems, async Rust/Tokio, graph traversal, serialization.

Cloud & Infrastructure: Docker, Kubernetes, AWS (Lambda, EC2, S3), GCP, GitHub Actions, CI/CD, Cargo, Babel, Yarn.

AI & ML: PyTorch, Hugging Face Transformers, ONNX, ONNX Runtime, TensorFlow/Keras, NumPy.

Relevant Experience

Oct 2025 – **TAHO (early-stage startup).**

Apr 2026 Product Engineer / Software Engineer - Rust, Python

- Built distributed inference infrastructure that enabled local offline execution of a 32B-parameter LLM on Mac hardware.
- Implemented streaming model execution components in Rust using ONNX Runtime, protobuf, and rkyv.
- Implemented graph traversal, tensor, and serialization components for model execution workflows.
- Integrated PyTorch, Hugging Face Transformers, and ONNX across model conversion, deployment, and runtime workflows.
- Developed graph visualization, debugging tools, and integration tests for inspecting model execution.

Aug 2021 - **Fuel Labs (early-stage startup).**

May 2023 Compiler Engineer - Rust

- Built core compiler components for Sway, a Rust-inspired smart contract language with a large developer ecosystem (61k+ [GitHub](#) stars).
- Implemented frontend compiler features across language semantics and type checking, including match expressions, traits, and generics.
- Built type-system infrastructure for name resolution, trait resolution, generic inference, and compiler diagnostics.
- Added compiler correctness checks, including pattern-match exhaustivity checking.
- Improved compiler errors, onboarding flows, and docs tooling from developer feedback.

May 2021 – **Google, Madison, WI.**

Aug 2021 Software Engineering Intern - C++

- Implemented load-balancing functionality in a high-performance networking library using RMA-based RPC mechanisms.
- Integrated low-level distributed systems components for communication with network hardware.

June 2018 - **Northeastern University, University of Massachusetts Amherst, Boston, MA, Amherst, MA.**

May 2021 Research Engineer - Rust, TypeScript, JavaScript, Python

- Built a serverless execution platform that compiled traced JavaScript workloads into Rust to improve latency and throughput.
- Designed compiler, runtime, networking, and storage systems for a full-stack distributed systems prototype.
- Developed distributed execution infrastructure using Docker, Kubernetes, Tokio, AWS Lambda, and Google Cloud Functions.
- Integrated database systems including CouchDB and Couchbase to support distributed execution workflows.
- Built debugging, observability, and testing infrastructure for distributed execution under limited visibility.

June 2017 – **National Aeronautics and Space Administration (NASA), Langley, VA.**

Aug 2017 Software Engineering Intern - C++

- Built and tested GPS-based geofencing systems for autonomous drones across heterogeneous hardware platforms.

Education

2021 **MS in Computer Science**, *The University of Massachusetts Amherst*, Amherst, MA.

2018 **BS in Computer Science**, *Trinity University*, San Antonio, TX.

2025 **Art & Design Intensive**, *Atelier de Sevres*, Paris, France.

Selected Publications

2021 Emily Herbert and Arjun Guha. A Language-based Serverless Function Accelerator. [[preprint](#), [repo](#)]

2020 Wang Cen, Emily A Herbert, and Peter J Haas. NIM: Modeling and Generation of Simulation Inputs via Generative Neural Networks. *Winter Simulation Conference*. [[paper](#)]. **Best Contributed Theoretical Paper Finalist.**